

A graph contrastive learning framework with multi-view adaptation for robust collaborative filtering

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HIGHLIGHTS

- MAGCL integrates multi-view contrastive learning into graph-based collaborative filtering.
- A high-order collaborative graph effectively captures richer interaction signals in sparse data.
- A trust score-based denoising mechanism filters unreliable interactions, enhancing robustness.
- Dynamic view fusion and cross-layer contrastive loss significantly improve generalization.
- A uniformity loss optimizes embedding quality, ensuring balanced representation distribution.

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ABSTRACT

Collaborative filtering is a core technique in recommender systems but suffers from sparse user-item interactions and noisy implicit feedback. We propose MAGCL, a Multi-view Adaptive Graph Contrastive Learning framework. It learns robust and informative user and item representations under sparse and noise-prone conditions. MAGCL constructs two complementary graph views: (i) a high-order collaborative graph that models user-user and item-item relations, and (ii) a trust-score-based denoised augmented graph that downweights unreliable interactions. An adaptive fusion strategy integrates information from multiple graph views. In addition, a joint objective combines multi-view contrastive loss, cross-layer contrastive loss, uniformity regularization, and Bayesian Personalized Ranking (BPR). This design improves both alignment and robustness. We evaluate MAGCL on two public datasets: MovieLens-1M, which is relatively dense and reliable, and Alibaba-iFashion, which is highly sparse and noisy. To ensure fair comparison, we adopt the same benchmarks and metrics as those used by nine representative baselines, including (1) matrix factorization or neural collaborative filtering methods (BPRMF, NeuMF), (2) graph collaborative filtering models (NGCF, LightGCN, MultiGCCF), and (3) contrastive or self-supervised recommenders (SGL, SimGCL, AdaMCL, ProtoAU). Under a unified evaluation protocol, MAGCL achieves the best overall performance on the Alibaba-iFashion dataset and competitive results on MovieLens-1M, where the strongest model ProtoAU performs slightly better because it leverages the denser user-item interactions. Specifically, compared with ProtoAU, MAGCL improves Recall@50 by 2.12%, NDCG@50 by 1.72%, and Recall@10 by 1.58% on the highly sparse Alibaba-iFashion dataset. These results demonstrate the particular effectiveness of MAGCL in recommendation scenarios characterized by sparse and noisy implicit feedback.

1. Introduction

Recommender systems aim to infer user preferences from historical user-item interactions, such as browsing, purchasing, and favoriting behaviors, to provide personalized services [1]. Collaborative filtering (CF) techniques exploit the collective behavior of user groups to predict individual preferences, enabling effective information filtering in

large-scale systems. In recent years, Graph Neural Networks (GNNs) have become prominent in recommender system research due to their ability to model complex graph-structured data. In CF scenarios, GNNs represent user-item interactions as bipartite graphs, where nodes represent users and items, and edges denote observed interactions. This formulation allows the recommendation task to be reframed as an edge

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prediction problem. By propagating messages across multiple layers, GNNs aggregate multi-hop neighbor information and capture high-order collaborative signals. This mechanism helps overcome limitations of traditional CF models in representing complex user-item relationships, thereby improving recommendation performance.

Despite notable progress, GNN-based collaborative filtering models still face key challenges. Particularly, user-item interaction data often suffer from sparsity and noise, directly degrading the quality of learned embeddings [2]. To address these issues, contrastive learning has emerged as a promising approach for uncovering latent patterns and enhancing model generalization. However, many methods build contrastive objectives only on the original user-item graph and do not explicitly model high-order user-user and item-item relationships. Prior studies have shown that high-order relationships contribute significantly to recommendation accuracy [3]. Although some studies [4] attempt to incorporate high-order information, their approaches are often indirect and lack precision. Moreover, the impact of high-order relationships varies across different nodes and may become redundant or even harmful in certain cases. As the number of GNN layers increases, high-order relationships in the user-user graph tend to become redundant. This redundancy weakens their original utility and may negatively impact model performance. To accurately evaluate the contribution of high-order relationships, it is necessary to develop a dynamic and fine-grained mechanism that adaptively regulates the receptive field of high-order signals. Recent studies further suggest that optimizing the uniformity of user and item representations can significantly improve model performance [5,6]. This motivates our exploration of uniformity optimization in MAGCL.

To address these challenges, we propose MAGCL, a multi-view adaptive graph contrastive learning framework. MAGCL explicitly captures high-order relationships between users and items by constructing high-order collaborative graphs, including user-user and item-item graphs. During user representation learning, MAGCL employs a dynamic fusion strategy to integrate embeddings from both the user-item interaction graph and the user-user graph. This enables the framework to better capture the role of higher-order relationships. Additionally, we introduce a trust-score-based denoising mechanism to construct a denoised augmented graph, which is then incorporated into a multi-view contrastive learning method. This method enables effective contrastive learning while mitigating noise interference. **Furthermore, MAGCL integrates cross-layer contrastive learning to alleviate noise introduced by aggregating high-order neighbors, and optimizes the uniformity loss function to further enhance recommendation performance.**

2. Related work

In recommender systems, GNNs have significantly advanced collaborative filtering by introducing novel mechanisms for modeling user-item interactions. Wang et al. [7] proposed NGCF, a framework that uses graph convolutional networks to capture high-order features in user-item interaction graphs. However, its intricate propagation mechanism constrained its expressive power. He et al. [8] showed that feature transformations and nonlinear activations in GCNs contributed little to collaborative filtering. They proposed LightGCN, a simplified architecture that achieved better performance and established a benchmark in the field. Sun et al. [9] developed MultiGCCF, which builds auxiliary graphs based on user-user and item-item similarities to enhance feature extraction. Peng et al. [10] proposed a graph denoising encoder that optimizes spectral feature extraction and improves training efficiency. Wang et al. [11] proposed DGCF, which leverages intent disentanglement to provide personalized recommendations. Liu et al. [12] introduced IMP-GCN, which tackles the problem of embedding similarity through subgraph partitioning. Fan et al. [13] proposed GraphDA, which refines the adjacency matrix to denoise data for active users and enrich representations for inactive ones. Despite these advancements,

data sparsity and noisy user feedback remain persistent challenges in collaborative filtering. Recent studies have begun addressing these issues from a noise-robust learning perspective. For example, Ye and Lu [14] investigated rating-flip noise and proposed a statistically consistent framework to improve recommendation robustness. Baldán et al. [15] developed a time-aware natural noise management model to handle temporal inconsistencies, while Zhang et al. [16] introduced a personalized denoising strategy for implicit-feedback data. In addition, Zhang et al. [17] analyzed adversarial collaborative filtering and demonstrated how perturbation-aware learning enhances robustness. Although auxiliary data from knowledge graphs [18–20] and social networks [21–23] can alleviate these issues, obtaining such data is often associated with considerable cost and effort.

To address data sparsity and noise in recommendation scenarios, contrastive learning methods have been increasingly adopted in recent years due to their ability to uncover latent data relationships and improve model generalization [24]. The core of contrastive learning lies in generating multi-view representations of the user-item interaction graph through data augmentation strategies and optimizing user and item embeddings via contrastive loss functions. SimGCL [25], XSimGCL [5], LightGCL [26], and SGCL [27] represent the recent state-of-the-art methods in contrastive learning and have demonstrated promising performance. Building upon these foundations, several recent studies have extended contrastive learning toward more robust and generalizable recommendation paradigms. Wang et al. [28] introduced high-order contrastive objectives to improve noise resilience in graph-based models, Zhang et al. [29] proposed a dual contrastive learning framework that jointly optimizes batch-level and feature-level consistency, Xie et al. [30] developed a knowledge-aware multi-view contrastive learning model that integrates user-item-entity semantics, and Dose et al. [31] designed an inductive contrastive framework capable of generalizing to unseen users and items. Despite these advances, many existing models remain vulnerable in complex environments due to their limited robustness to noise and poor capacity for modeling high-order relationships.

3. Notation and preliminaries

We define the user set and item set as $\mathcal{U}$ and $\mathcal{I}$, respectively, where $|\mathcal{U}| = m$ and $|\mathcal{I}| = n$. Let G denote the user-item interaction graph, with its adjacency matrix $E^G \in \mathbb{R}^{(m+n) \times (m+n)}$. The user-item interaction matrix $R \in \{0, 1\}^{m \times n}$ encodes whether user u has interacted with item i , where $r_{u,i} = 1$ if an interaction exists, and $r_{u,i} = 0$ otherwise. Let $\check{G}$ denote the high-order collaborative graph, whose adjacency matrix is $E^{\check{G}} \in \mathbb{R}^{(m+n) \times (m+n)}$. It comprises the user-user interaction matrix $\mathcal{P} \in \mathbb{R}^{m \times m}$ and the item-item interaction matrix $\mathcal{Q} \in \mathbb{R}^{n \times n}$. Let $\check{G}$ denote the denoised augmented graph, with its adjacency matrix $E^{\check{G}} \in \mathbb{R}^{(m+n) \times (m+n)}$. **Let $\check{R}$ denote the user-item interaction matrix of the denoised augmented graph**, where $\check{r}_{u,i}$ represents the denoised interaction weight between user u and item i . The formal definitions of E^G , $E^{\check{G}}$, and $E^{\check{G}}$ are provided as follows:

$$E^G = \begin{pmatrix} 0 & R \\ R^T & 0 \end{pmatrix}; E^{\check{G}} = \begin{pmatrix} \mathcal{P} & 0 \\ 0 & \mathcal{Q} \end{pmatrix}; E^{\check{G}} = \begin{pmatrix} 0 & \check{R} \\ \check{R}^T & 0 \end{pmatrix} \quad (1)$$

Let $N_{G(x)}$, $N_{\check{G}(x)}$, and $N_{\check{G}(x)}$ denote the one-hop neighborhoods of node x in the user-item interaction graph G , the denoised augmented graph $\check{G}$, and the high-order collaborative graph $\check{G}$, respectively. Throughout, $\mathbb{R}$ denotes the set of real numbers, and $|\cdot|$ denotes the cardinality of a set.

4. MAGCL framework

The architecture of MAGCL, as shown in Fig. 1, consists of three modules: (i) high-order collaborative and denoised augmented graph construction, (ii) multi-graph node representation learning, and (iii) joint learning.

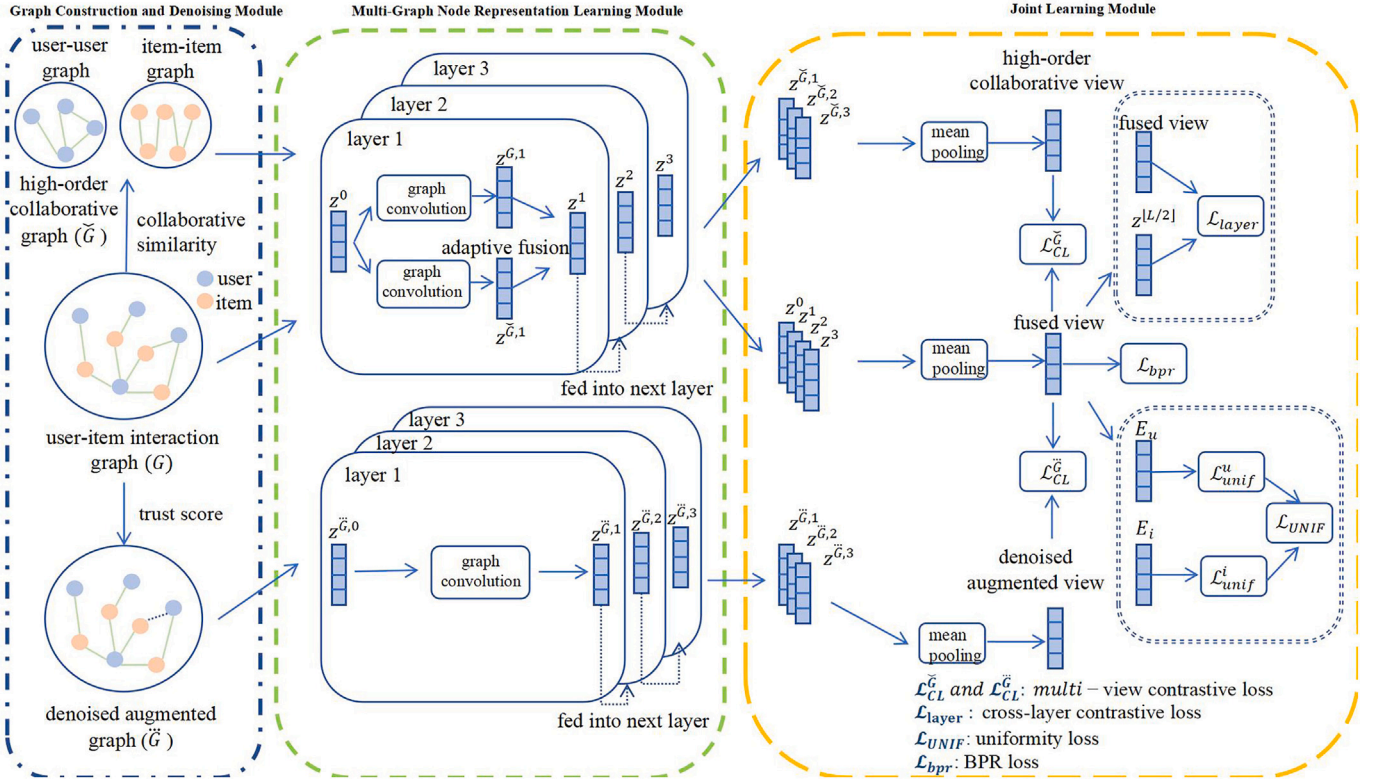


Fig. 1. MAGCL framework architecture.

4.1. Construction of high-order and denoised augmented graphs

4.1.1. High-order collaborative graph construction

In recommender systems, explicitly modeling high-order relationships between users and items can improve the accuracy of recommendations. To capture such relationships more effectively, we construct a high-order collaborative graph modeling both user-user and item-item high-order relations.

To capture user preference similarities and latent item associations, we define collaborative preference users and collaboratively associated items. Collaborative preference users refer to users who exhibit similar preferences inferred from historical interactions, while collaboratively associated items denote items that are co-interacted with by the same users. To measure the strength of these relationships, we employ collaborative similarity to assess the closeness between collaborative preference users or collaboratively associated items. Collaborative similarity is typically computed based on the overlap in interaction histories between users or between items [32]. In this work, collaborative similarity is computed using the Jaccard coefficient [33], based on the user-item interaction graph, as follows:

$$JSC_{x,y} = \frac{|N_{G(x)} \cap N_{G(y)}|}{|N_{G(x)} \cup N_{G(y)}|} \quad (2)$$

When computing collaborative similarity, x and y represent two distinct users in G if user similarity is measured, or two distinct items if item similarity is measured. Based on the computed collaborative similarity, the adjacency matrix of the high-order collaborative graph is constructed as follows:

$$E_{x,y}^{\tilde{G}} = \begin{cases} JSC_{x,y} & \text{if } JSC_{x,y} \geq \beta \text{ or top } S \text{ values for node } x \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

Here, $E_{x,y}^{\tilde{G}}$ represents the interaction weight between nodes x and y in the adjacency matrix of the high-order collaborative graph, and β is a hyperparameter. When $JSC_{x,y}$ is greater than or equal to β , $E_{x,y}^{\tilde{G}}$ is assigned the value of $JSC_{x,y}$. Additionally, S is another hyperparameter. For each node x , even if the $JSC_{x,y}$ of certain nodes y is less than β , $E_{x,y}^{\tilde{G}}$ is still assigned the value of $JSC_{x,y}$ as long as these nodes belong to the top- S nodes with the highest collaborative similarity to x . This setup ensures that each node maintains connections with at least a certain number of neighbor nodes, thereby effectively preserving key similarity relationships.

4.1.2. Denoised augmented graph construction

Noisy user-item interactions degrade performance and distort preference modeling. To address this, we leverage graph topology to construct a denoised and augmented user-item graph.

Definition. The trust score-based denoising mechanism quantifies the reliability of each user-item interaction through a trust score $t_{u,i}$, which measures the consistency between the embeddings of user u and item i across multi-hop neighborhoods. Interactions whose local structures are inconsistent with their higher-order neighbors are assigned smaller $t_{u,i}$ values. During graph convolution, $t_{u,i}$ acts as an edge weight to suppress unreliable interactions, resulting in a denoised and augmented graph that emphasizes trustworthy links. This process effectively reduces the impact of noisy implicit feedback in large-scale recommender systems.

During graph convolution, the trust score acts as an edge weight to suppress noisy or unreliable interactions, leading to a denoised and augmented graph that emphasizes trustworthy links. This denoising mechanism effectively mitigates the impact of noisy implicit feedback in large-scale recommender systems. This refined structure supports contrastive-learning objectives, providing reliable training signals and improving robustness to noise.

To construct the denoised augmented graph, trust scores are first computed for user-item interactions, based on which noisy data are

identified and removed. To achieve this, we leverage homophily theory [34], which suggests that similar entities are more likely to establish connections. In other words, the likelihood of interaction increases with the similarity between users and items.

Noise can interfere with the direct computation of similarity from individual embeddings, leading to instability. To mitigate this, we adopt a neighborhood aggregation strategy to improve the reliability of similarity estimation.

Specifically, we compute a node's embedding vector based on its one-hop neighbor information, rather than directly using the node's own embedding vector as input for trust score computation. This strategy is motivated by the observation that, under noise interference, an individual node's embedding vector may exhibit notable bias, whereas the collective information of its neighbors can more comprehensively reflect the true characteristics of users and items.

Given a user u , its representation vector is obtained by the weighted sum of the embedding vectors of its interacted neighbor items. Similarly, for an item i , its representation vector is derived from the weighted sum of the embedding vectors of its interacted neighbor users. The specific computation is as follows:

$$z_u^{G,1} = \sum_{i \in N_{G(u)}} \frac{1}{\sqrt{|N_{G(u)}| \times |N_{G(i)}|}} z_i^{G,0} \quad (4)$$

$$z_i^{G,1} = \sum_{u \in N_{G(i)}} \frac{1}{\sqrt{|N_{G(i)}| \times |N_{G(u)}|}} z_u^{G,0} \quad (5)$$

Here, $z_u^{G,0}$ and $z_i^{G,0}$ represent the initial embedding vectors of user u and item i in the user-item interaction graph G , respectively. The term $\frac{1}{\sqrt{|N_{G(u)}| \times |N_{G(i)}|}}$ is a symmetric normalization factor used to balance the influence of nodes with different degrees, preventing high-degree nodes from disproportionately dominating the aggregation result. We adopt cosine similarity to compute the trust score of the interaction between user u and item i , with the formula given as follows:

$$\cos(z_u^{G,1}, z_i^{G,1}) = \frac{(z_u^{G,1})^T z_i^{G,1}}{\|z_u^{G,1}\|_2 \cdot \|z_i^{G,1}\|_2} \quad (6)$$

Here, $\|\cdot\|_2$ denotes the Euclidean (ℓ_2) norm, and $\cos(\cdot, \cdot)$ denotes the cosine similarity.

Since the cosine similarity in Eq. (6) may yield negative values, a numerical mapping is applied to ensure that the trust scores lie within the range $[0, 1]$, as follows:

$$T_{u,i} = \left(\cos(z_u^{G,1}, z_i^{G,1}) + 1 \right) / 2 \quad (7)$$

Here, $T_{u,i}$ denotes the trust score of the interaction between user u and item i , with higher values indicating more reliable interactions. Based on these scores, we adopt both hard and soft denoising strategies to construct the denoised augmented graph, as follows:

$$\tilde{r}_{u,i} = \begin{cases} T_{u,i} & \text{if } T_{u,i} > \theta \\ 0 & \text{if } T_{u,i} \leq \theta \end{cases} \quad (8)$$

Here, θ is a denoising threshold hyperparameter. When the trust score $T_{u,i}$ is less than or equal to θ , the interaction is considered noisy, and the hard denoising strategy assigns the denoised interaction weight $\tilde{r}_{u,i}$ a value of 0. When $T_{u,i} > \theta$, the soft denoising strategy assigns the denoised interaction weight a value of $T_{u,i}$.

4.2. Multi-graph node representation learning

After constructing the user-item interaction graph G , the denoised augmented graph $\tilde{G}$, and the high-order collaborative graph $\check{G}$, we perform graph convolution on each of them to learn node embedding

representations. To better capture high-order information, we introduce a dynamic fusion strategy in the convolution over G and $\check{G}$, which adaptively combines information from both graphs to produce more expressive node representations for recommendation.

4.2.1. Multi-graph convolution

For a node x , we denote its embedding vectors at the l -th layer in the user-item interaction graph and the high-order collaborative graph as $z_x^{G,l}$ and $z_x^{\check{G},l}$, respectively. Following the design of LightGCN [8], we eliminate redundant feature transformations and nonlinear activations in the graph convolution process and retain only the neighborhood aggregation function. The graph convolution operation in the user-item interaction graph is defined as follows:

$$z_x^{G,l} = \sum_{y \in N_G(x)} \frac{1}{\sqrt{|N_G(x)| |N_G(y)|}} z_y^{G,l-1} \quad (9)$$

This formula updates the embedding vector of node x at the l -th layer in the user-item interaction graph by aggregating information from its one-hop neighbors. The term $\frac{1}{\sqrt{|N_G(x)| |N_G(y)|}}$ is a symmetric normalization factor used to balance the influence of nodes with varying degrees. We denote by L the total number of propagation layers in the encoder; layer indices satisfy $l \in \{1, \dots, L\}$. The graph convolution operation in the high-order collaborative graph is defined as follows:

$$z_x^{\check{G},l} = \sum_{y \in N_{\check{G}}(x)} \frac{1}{|N_{\check{G}}(x)|} z_y^{\check{G},l-1} \quad (10)$$

Here, $\frac{1}{|N_{\check{G}}(x)|}$ represents a symmetric normalization factor. This formula updates the embedding vector of node x at the l -th layer in the high-order collaborative graph by aggregating information from its neighbors, similar to the update in the user-item interaction graph G . However, unlike G , which relies on direct user-item interaction relationships, the neighbors in $\check{G}$ are derived from high-order relationships that capture user-user and item-item connections, which provide richer feature representations.

The graph convolution operation in the denoised augmented graph is defined as follows:

$$z_u^{\tilde{G},l} = \sum_{i \in N_{\tilde{G}}(u)} \frac{\tilde{r}_{u,i}}{\sum_{j \in N_{\tilde{G}}(u)} \tilde{r}_{u,j}} z_i^{\tilde{G},l-1} \quad (11)$$

$$z_i^{\tilde{G},l} = \sum_{u \in N_{\tilde{G}}(i)} \frac{\tilde{r}_{u,i}}{\sum_{j \in N_{\tilde{G}}(i)} \tilde{r}_{j,i}} z_u^{\tilde{G},l-1} \quad (12)$$

Here, $z_u^{\tilde{G},l}$ and $z_i^{\tilde{G},l}$ represent the embedding vectors of user u and item i at the l -th layer in the denoised augmented graph. $\tilde{r}_{u,i}$ denotes the denoised interaction weight between user u and item i . The normalization terms $\frac{\tilde{r}_{u,i}}{\sum_{j \in N_{\tilde{G}}(u)} \tilde{r}_{u,j}}$ and $\frac{\tilde{r}_{u,i}}{\sum_{j \in N_{\tilde{G}}(i)} \tilde{r}_{j,i}}$ adjust the interaction weights based on the neighbors, thereby mitigating biases caused by differences in neighbor counts or imbalanced weight distributions. $z_u^{G,0}$ and $z_i^{G,0}$ are the initial embedding vectors of the user and item, respectively.

4.2.2. Adaptive embedding fusion

To adaptively capture interaction information from the user-item interaction graph G , user-user and item-item relationships from the high-order collaborative graph $\check{G}$, we employ a dynamic fusion strategy to combine $z_x^{G,l}$ and $z_x^{\check{G},l}$. This strategy optimizes user-user high-order relationships but is applicable to all nodes, including both users and items. The dynamic fusion strategy is implemented as follows:

$$z_x^l = z_x^{G,l} + \eta_x z_x^{\check{G},l} \quad (13)$$

$$\eta_x = \frac{\gamma}{c(z_x^{G,l}, z_x^{\check{G},l}) d_x + l} \quad (14)$$

$$d_x = \frac{\log(|N_{G(x)}|)}{\frac{1}{|V|} \sum_{v \in V} \log(|N_{G(v)}|)} \quad (15)$$

Here, z_x^l represents the fused embedding vector of node x at the l -th layer, and serves as the input to the next graph convolution layer. z_x^0 represents the initial embedding vector of node x . $z_x^{G,l}$ is the embedding vector of node x at the l -th layer in the user-item interaction graph G , and $z_x^{\tilde{G},l}$ denotes the embedding vector of node x at the l -th layer in the high-order collaborative graph $\tilde{G}$. V denotes the set of all nodes, and γ is a hyperparameter controlling the weight of $z_x^{G,l}$. The parameter l represents the current graph convolution layer. As the number of layers increases, the receptive field becomes broader, and the contribution of high-order relationships may decrease or weaken or even turn negative. Therefore, l and η_x are designed to vary inversely. d_x represents the normalized user activity in the user-item interaction graph. To mitigate excessive differences in user activity, we apply logarithmic processing. For highly active users, one-hop neighbor information is typically sufficient, reducing the need for high-order information. Thus, the weight of high-order relationships should decrease as d_x increases. The function $c(z_x^{G,l}, z_x^{\tilde{G},l})$ is the cosine similarity function used to compute the similarity between $z_x^{G,l}$ and $z_x^{\tilde{G},l}$. We adopt the cosine similarity $c(a, b) = \frac{a^T b}{\|a\|_2 \|b\|_2}$, and V denotes the set of all nodes. The activity term d_x normalizes user activity based on node degrees as defined in Eq. (15). When the similarity is high, high-order information may become redundant, and η_x should be reduced to avoid excessive fusion.

Following [10], we observe that low-activity users tend to have fewer interactions. As a result, high-order information extracted from the collaborative graph $\tilde{G}$ is more valuable to them. This implies a higher tolerance for redundant high-order relationships among low-activity users. In other words, the similarity of embedding vectors is weighted based on user activity in the user-item interaction graph G . This strategy dynamically adjusts the weight of high-order information, enabling more accurate modeling of its impact.

4.2.3. Layer-wise fusion and prediction function

After l layers of graph convolution, we use average pooling to fuse the embedding vectors from all convolution layers, such as z_u^l and z_i^l (where $l = 1, \dots, L$). We also fuse the initial embedding vectors, such as z_u^0 and z_i^0 , to obtain the final embedding vectors. These are computed as follows:

$$e_u = \frac{1}{L+1} \sum_{l=0}^L z_u^l \quad (16)$$

$$e_i = \frac{1}{L+1} \sum_{l=0}^L z_i^l \quad (17)$$

Here, e_u and e_i represent the final embedding vectors of user u and item i , respectively.

After obtaining the final embedding vectors for user u and item i , we compute the predicted rating of user u for item i . This is calculated as the inner product of their embedding vectors:

$$\hat{y}_{u,i} = e_u^T e_i \quad (18)$$

Here, $\hat{y}_{u,i}$ denotes the predicted rating for the interaction between user u and item i in the framework.

For the recommendation task, we use the Bayesian Personalized Ranking (BPR) loss function to optimize the ranking. It is defined as follows:

$$\mathcal{L}_{BPR} = \sum_{(u,i,j) \in \mathcal{Y}} -\log \sigma(\hat{y}_{u,i} - \hat{y}_{u,j}) \quad (19)$$

Here, $\mathcal{Y}$ is defined as $\{(u, i, j) \mid u \in \mathcal{U}, i \in \mathcal{I}, r_{u,i} \neq 0, r_{u,j} = 0\}$, where σ denotes the Sigmoid activation function.

The BPR loss function aims to maximize the difference between the predicted scores of positive samples (items with interactions) and negative samples (items without interactions). By doing so, it optimizes the ranking performance of the framework by ensuring that positive samples receive substantially higher scores than negative ones.

4.3. Joint learning

We further introduce three key objectives: the cross-view, cross-layer, and uniformity contrastive losses, which jointly guide the model to learn robust and diverse representations. Specifically, the cross-view contrastive loss encourages consistency across different graph views, the cross-layer loss preserves information across propagation depths, and the uniformity loss promotes diversity in the embedding space.

4.3.1. Multi-view contrastive learning

The embedding vectors obtained from the high-order collaborative graph and the denoised augmented graph are referred to as the high-order collaborative view and the denoised augmented view, respectively.

The final fused embeddings are denoted as $e_v (v \in \mathcal{U} \cup \mathcal{I})$, and referred to as the fused view. These embeddings are derived from Eqs. (16) and (17) and are used to construct subsequent contrastive learning tasks.

Using average pooling, we fuse the embedding vectors from all convolution layers of the high-order collaborative graph, as given in Eq. (10) (e.g., $z_v^{G,l}$, for $l = 1, \dots, L$). The final embedding vector of the high-order collaborative graph, $e_v^{\tilde{G}}$, is then obtained, which is regarded as the high-order collaborative view.

$$e_v^{\tilde{G}} = \frac{1}{L} \sum_{l=1}^L z_v^{G,l} \quad (20)$$

Similarly, using average pooling, we fuse the embedding vectors from all convolutional layers of the denoised augmented graph (e.g., $z_u^{\tilde{G},l}$ and $z_i^{\tilde{G},l}$, for $l = 1, \dots, L$). The fusion is computed as follows:

$$e_u^{\tilde{G}} = \frac{1}{L} \sum_{l=1}^L z_u^{\tilde{G},l} \quad (21)$$

$$e_i^{\tilde{G}} = \frac{1}{L} \sum_{l=1}^L z_i^{\tilde{G},l} \quad (22)$$

Here, $e_u^{\tilde{G}}$ and $e_i^{\tilde{G}}$ denote the final embedding vectors for user u and item i in the denoised augmented graph, respectively. Subsequently, we use $e_v^{\tilde{G}}$ to represent both $e_u^{\tilde{G}}$ and $e_i^{\tilde{G}}$, i.e., the final embedding vector of the denoised augmented graph, which we treat as the denoised augmented view.

The fused view integrates information from the user-item interaction graph and the high-order collaborative graph to generate richer feature representations. To exploit this advantage, the model performs multi-view contrastive learning, where the fused view is contrasted with the denoised-augmented and high-order collaborative views. Corresponding denoising and high-order contrastive losses are then calculated as follows:

$$\mathcal{L}_{CL}^{\tilde{G}} = \sum_{v \in \mathcal{B}} -\log \frac{\exp((e_v^T e_v^{\tilde{G}})/\tau)}{\sum_{x \in \mathcal{B}} \exp((e_v^T e_x^{\tilde{G}})/\tau)} \quad (23)$$

$$\mathcal{L}_{CL}^{\tilde{G}} = \sum_{v \in \mathcal{B}} -\log \frac{\exp((e_v^T e_v^{\tilde{G}})/\tau)}{\sum_{x \in \mathcal{B}} \exp((e_v^T e_x^{\tilde{G}})/\tau)} \quad (24)$$

$$\mathcal{L}_{CL} = \mathcal{L}_{CL}^{\tilde{G}} + \alpha \mathcal{L}_{CL}^{\tilde{G}} \quad (25)$$

Here, $\mathcal{B}$ denotes the set of users or positive sample items within the current mini-batch used to form contrastive pairs. τ is the temperature coefficient, and $\exp(\cdot)$ is the exponential function. For example, in

$\mathcal{L}_{CL}^G$, e_v and e_v^G represent the embedding vectors of node x in the fused view and the denoised augmented view, respectively, with their inner product measuring the vector similarity. $\mathcal{L}_{CL}^G$ aims to maximize mutual information between the fused view and the denoised augmented view. Similarly, $\mathcal{L}_{CL}^G$ strengthens the alignment between the fused view and the high-order collaborative view. α is a hyperparameter that balances the contribution of high-order contrastive loss, and $\mathcal{L}_{CL}$ is the multi-view contrastive loss, defined as the weighted sum of $\mathcal{L}_{CL}^G$ and $\mathcal{L}_{CL}^G$.

4.3.2. Cross-layer contrastive learning

As the number of graph convolutional layers increases, the receptive field of the fused view expands exponentially, which leads to a much larger receptive field compared to traditional GNN-based frameworks and may introduce noise into the representations. For instance, when the number of convolution layers is $L/2$, the receptive field of the fused view is equivalent to that of L layers in a traditional GNN. When referencing the middle layer, we use $z_v^{[L/2]}$ to avoid ambiguity for odd L . To mitigate noise caused by the aggregation of high-order neighbors, we introduce a cross-layer contrastive loss and construct contrastive views based on the fused view e_v and the intermediate layer embedding vectors $z_v^{[L/2]}$. The cross-layer contrastive loss is defined as:

Definition. The *cross-layer contrastive loss* aims to align embeddings from different propagation depths, enforcing semantic consistency between the top-layer and intermediate-layer representations of each node. This objective mitigates the over-smoothing effect introduced by high-order aggregation, ensuring that deeper layers preserve discriminative information learned in shallower layers.

$$\mathcal{L}_{\text{layer}} = \sum_{v \in B} -\log \frac{\exp \left(\left(e_v^T \cdot z_v^{[L/2]} \right) / \tau \right)}{\sum_{x \in B} \exp \left(\left(e_v^T \cdot z_x^{[L/2]} \right) / \tau \right)} \quad (26)$$

4.3.3. Uniformity adjustment

Uniformity is a key attribute in obtaining high-quality representations in contrastive learning, typically defined as the logarithmic form of the expected Gaussian potential. Recent studies [35] emphasize optimizing both uniformity and alignment during training, while [5,25] highlight the particularly critical role of uniformity. Literature [6] notes that BPR enhances alignment through negative sampling but is less effective at optimizing uniformity. Improving uniformity further can compensate for BPR's limitations in representation learning, and thus enhance framework generalization and recommendation performance. Moreover, uneven embedding distributions may amplify noise, making the framework more sensitive to outliers. Therefore, we regulate the uniformity of fused embedding vectors for users and items within a batch, with the goal of achieving an even distribution on the unit hypersphere, so as to enhance both the robustness of the representations and the performance of the framework. Given the potential differences in data distributions between users and items, it is more appropriate to measure uniformity separately for each.

Specifically, we sample user pairs (u, u') from the deduplicated user distribution. Their embedding vectors are normalized using L2 normalization to obtain $f(u)$ and $f(u')$. The Gaussian potential between them is then computed to reflect their proximity in the embedding space. The average Gaussian potential across all possible user pairs is computed. A high average indicates an uneven distribution (i.e., many embeddings clustered together), while a low average suggests a uniform distribution. For optimization, we take the natural logarithm of the average potential, converting it into a loss function to obtain the uniformity.

Definition. The *uniformity loss* quantifies how evenly the normalized embeddings are distributed on the unit hypersphere. It minimizes the expected Gaussian potential among distinct nodes, thereby preventing representation collapse and improving feature diversity. Formally, it is computed as:

$$\mathcal{L}_{\text{unif}}^u = \log \mathbb{E}_{u, u' \sim \text{Uniq}(P_{\text{user}})} e^{-2\|f(u) - f(u')\|^2} \quad (27)$$

Here, $\text{Uniq}(\cdot)$ denotes the deduplication operation, $\mathbb{E}_{u, u' \sim \text{Uniq}(P_{\text{user}})}$ averages the Gaussian potential across all user pairs, and $\log(\cdot)$ is the natural logarithm. In this context, $f(\cdot)$ represents the L2 normalization function that maps embeddings to the unit hypersphere. P_{user} represents the user node distribution. The uniformity loss for items follows a similar approach:

$$\mathcal{L}_{\text{unif}}^i = \log \mathbb{E}_{i, i' \sim \text{Uniq}(P_{\text{item}})} e^{-2\|f(i) - f(i')\|^2} \quad (28)$$

The item-side uniformity $\mathcal{L}_{\text{unif}}^i$ is defined analogously using distinct item pairs sampled from $\text{Uniq}(P_{\text{item}})$.

By combining the aforementioned two losses, we obtain the total uniformity loss:

$$\mathcal{L}_{\text{UNIF}} = \frac{1}{2} \left(\mathcal{L}_{\text{unif}}^u + \mathcal{L}_{\text{unif}}^i \right) \quad (29)$$

4.3.4. Multi-task training

To further enhance the quality of node representations, we adopt a multi-task training strategy that jointly optimizes the loss functions mentioned above. The final loss function includes BPR loss, multi-view contrastive loss, uniformity loss, and cross-layer contrastive loss, as defined below:

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \lambda_1 \mathcal{L}_{\text{CL}} + \lambda_2 \mathcal{L}_{\text{layer}} + \lambda_3 \mathcal{L}_{\text{UNIF}} + \lambda_4 \|\Theta\|_2 \quad (30)$$

Here, λ_1 , λ_2 , λ_3 , and λ_4 are hyperparameters that control the weights of the multi-view contrastive loss, cross-layer contrastive loss, uniformity loss, and regularization term, respectively. Additionally, Θ represents the set of all parameters in the framework.

5. Comprehensive experimental study

5.1. Datasets

To assess the effectiveness of the proposed MAGCL framework, we conduct experiments on two publicly available datasets: MovieLens-1M [4] and Alibaba-iFashion [36]. MovieLens-1M is a movie dataset that records user ratings for films, while Alibaba-iFashion is a highly sparse dataset that contains extensive records of user interactions with fashion items. Detailed information about the datasets can be found in Table 1.

For these datasets, the interaction data are randomly split into training, validation, and test sets in an 8:1:1 ratio.

We evaluate MAGCL on two widely used public datasets, MovieLens-1M and Alibaba-iFashion, which were selected as complementary benchmarks representing two major regimes in practical recommender systems. MovieLens-1M is relatively dense and behaviorally reliable. It is suitable for testing whether MAGCL leverages high-order signals and multi-view contrastive objectives to improve recommendation accuracy. By contrast, Alibaba-iFashion is extremely sparse (interaction density $\approx 7 \times 10^{-5}$) and contains noisy implicit feedback from large-scale online browsing. This dataset directly reflects the conditions that MAGCL is designed to address, namely sparse and noise-prone implicit feedback, where the proposed trust score based denoising and adaptive multi-graph fusion are particularly effective. Together, the two datasets cover dense and reliable interactions as well as highly sparse and noisy scenarios, enabling a comprehensive evaluation. Moreover, all nine baseline models compared in this work were originally evaluated on the same two datasets and metrics (Recall@K and NDCG@K), which ensures fair and reproducible comparisons.

Table 1
Overview of datasets.

Dataset	Users	Items	Interactions	Density
MovieLens-1M	6040	3629	836,478	0.03816
Alibaba-iFashion	300,000	81,614	1,607,813	0.00007

5.2. Hyperparameter tuning and sensitivity analysis

In this paper, the MAGCL framework and all baseline models are implemented using RecBole [37], **a unified open-source framework designed for the development and reproduction of recommender systems.** In constructing the high-order collaborative graph, the hyperparameter S is fixed at 5, and θ is set to 0.8, as defined in Eq. (3).

To ensure fair comparisons, all frameworks are configured with an embedding dimension of 64 and a batch size of 4096. The Adam optimizer is used for training, with parameters initialized via the Xavier method. To prevent overfitting, an early stopping strategy is employed, which halts training if the validation metric (e.g., Recall@20) does not improve for 10 consecutive epochs. The hyperparameters $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are tuned within the range [1e-9, 1e-3], and the denoising hyperparameter θ is optimized within [0.02, 0.1].

5.2.1. Hyperparameter tuning of τ , γ , and α

This section analyzes the impact of three key hyperparameters in the MAGCL framework on the MovieLens-1M and Alibaba-iFashion datasets.

In contrastive learning, the temperature coefficient τ is an important hyperparameter that regulates the similarity distribution between samples, influencing the framework's ability to differentiate between positive and negative samples. To investigate its impact on MAGCL performance, we experiment with τ in the range [0.05, 0.25] and evaluate Recall@20 and NDCG@20 on both datasets. The results are illustrated in Fig. 2.

The results reveal that, for both datasets, the model performance first improves and subsequently declines as τ increases. On MovieLens-1M, MAGCL achieves its best performance at $\tau = 0.1$, whereas on Alibaba-iFashion, the best performance occurs at $\tau = 0.2$. Further analysis shows that a high τ leads to a flatter similarity distribution, which diminishes the model's ability to effectively distinguish between positive and negative samples. Conversely, a low τ sharpens the distribution, which may reduce the model's sensitivity to low-similarity samples and thereby degrade the overall recommendation quality. Therefore, a moderate value for τ is crucial for enhancing the performance of contrastive learning and recommendation tasks.

In Eq. (14), γ controls the contribution of higher-order relationships in dynamic fusion. To examine its influence, we vary γ within the range [0.1, 1.3] and evaluate its effect using Recall@20 and NDCG@20 on both datasets. The results are illustrated in Fig. 3.

The performance on both datasets exhibits a non-monotonic trend, initially improving and then declining as γ increases. MAGCL achieves its best performance at $\gamma = 1.0$ on MovieLens-1M and at $\gamma = 0.4$ on Alibaba-iFashion. A small γ fails to fully exploit high-order relationship modeling, limiting the ability to capture deep user-item dependencies. In contrast, a large γ may introduce redundancy or noise, thereby reducing discriminative power and overall performance.

The hyperparameter α in Eq. (25) determines the trade-off between the denoising contrastive loss and the high-order contrastive loss. To examine its influence, we vary α within the range [0.5, 1.5] and evaluate the results using Recall@20 and NDCG@20. The results are presented in Fig. 4.

The performance on both datasets exhibits a non-monotonic trend, initially improving and subsequently declining as α increases. MAGCL achieves its best performance at $\alpha = 0.7$ on MovieLens-1M, and at $\alpha = 0.9$ on Alibaba-iFashion. These results indicate that the optimal balance between denoising and high-order contrastive losses is dataset-dependent. A higher α emphasizes the high-order contrastive loss, which may reduce robustness to noise, whereas a lower α prioritizes the denoising contrastive loss, potentially leading to underutilization of high-order relational signals.

5.2.2. Performance sensitivity to the number of GNN layers

To investigate the impact of the number of GNN layers L on MAGCL performance, we conduct experiments on MovieLens-1M and Alibaba-iFashion, testing L in the range [1, 4], using Recall@20 and NDCG@20 as evaluation metrics. The results are shown in Fig. 5.

Experimental results indicate that MAGCL's performance initially improves as the number of GNN layers increases, reaching an optimum at $L = 3$, beyond which it begins to decline due to over-smoothing effects. When L increases from 1 to 3, the model performance steadily improves. Specifically, the model performs suboptimally at $L = 1$. At $L = 2$, the model captures richer high-order connectivity information from the graph. This improves recommendation accuracy. Increasing the depth to $L = 3$ further enhances the model's ability to aggregate and propagate information. As a result, it learns more comprehensive representations for users and items and achieves the best performance. These results show that increasing the number of GNN layers within a reasonable range can help capture richer interaction patterns and improve recommendation accuracy. However, performance drops from

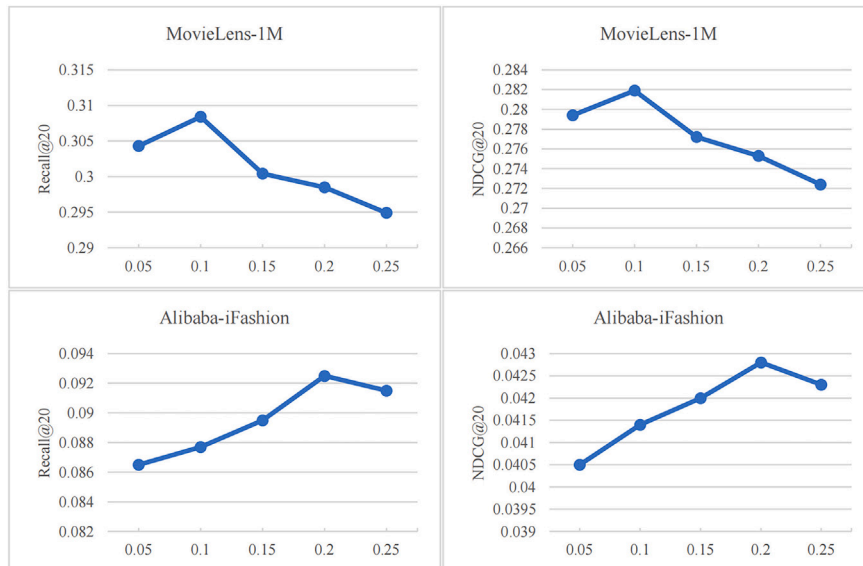


Fig. 2. Effect of temperature coefficient τ on recommendation performance.

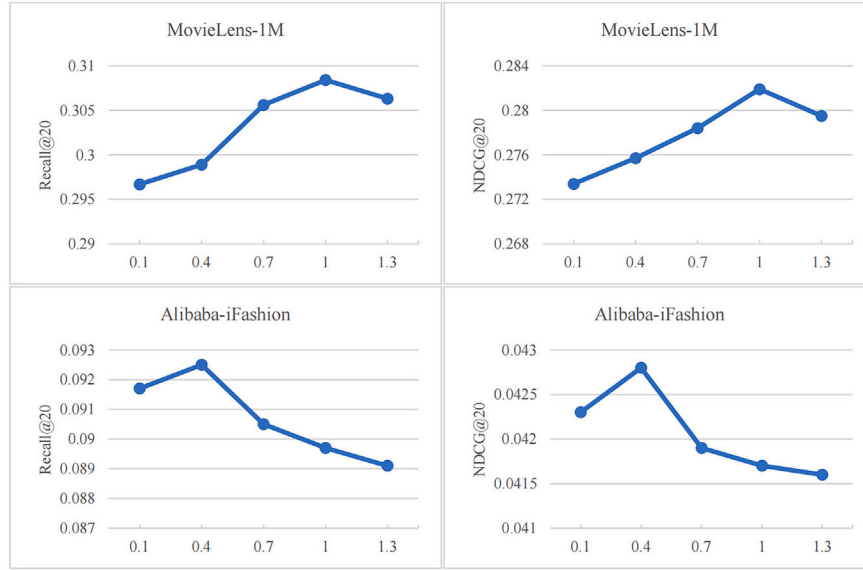


Fig. 3. Effect of hyperparameter γ on recommendation performance.

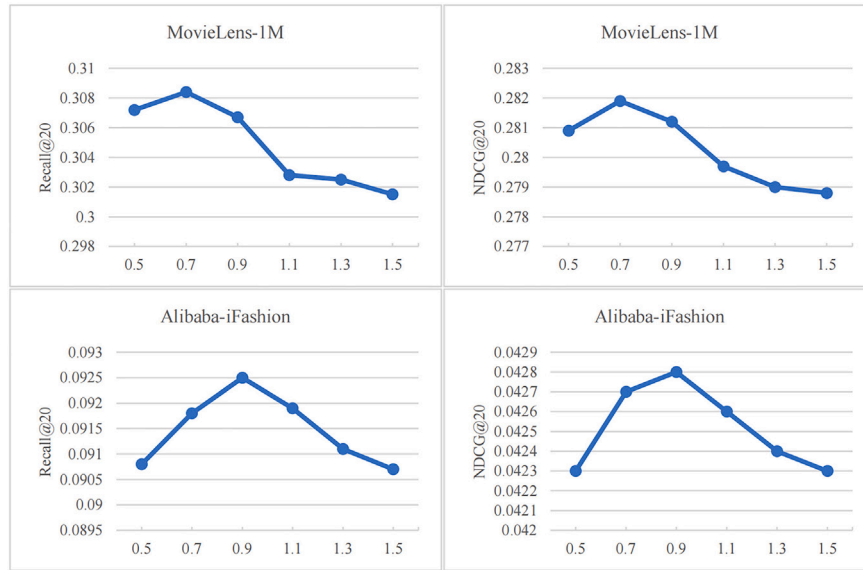


Fig. 4. Effect of hyperparameter α on recommendation performance.

$L = 3$ to $L = 4$, likely due to over-smoothing. As depth increases, node representations become increasingly similar. This weakens the model's ability to distinguish individual differences and lowers recommendation quality.

5.3. Comparative experiments

To thoroughly evaluate the effectiveness of the proposed MAGCL framework, we compare MAGCL with nine representative recommendation models, which can be grouped into three major categories. (1) Matrix Factorization or Neural Collaborative Filtering Methods: BPRMF and NeuMF, which model user-item interactions through latent factorization or nonlinear neural network structures. (2) Graph Collaborative Filtering Models: NGCF, LightGCN, and MultiGCCF, which enhance representation learning by propagating high-order connectivity along user-item bipartite graphs. (3) Contrastive or Self-Supervised

Learning Recommenders: SGL, SimGCL, AdaMCL, and ProtoAU, which employ self-supervised contrastive objectives to improve robustness and alleviate data sparsity. All baseline models were evaluated under identical experimental settings using the RecBole framework, with consistent embedding size, optimizer, and evaluation metrics (Recall@K and NDCG@K), ensuring fair and reproducible comparisons.

- **BPRMF** [38]: A classical matrix factorization framework that uses Bayesian personalized ranking loss to learn latent user-item representations.
- **NeuMF** [39]: A neural network-based recommendation framework that replaces traditional matrix factorization dot products with multilayer perceptrons to enhance expressiveness.
- **NGCF** [7]: Leverages GNN to model user-item interaction graphs, updating node representations through message passing and aggregation to capture high-order relationships.

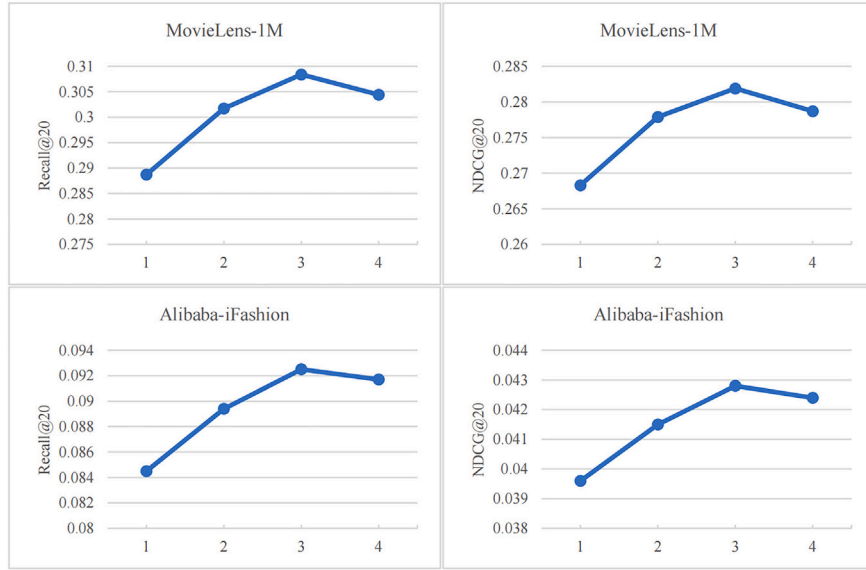


Fig. 5. Performance variation with respect to GNN layer depth L .

Table 2

Comparison of methods on MovieLens-1M and Alibaba-iFashion.

Datasets	Metric	BPRMF	NeuMF	NGCF	MultiGCCF	LightGCN	SimGCL	SGL	AdaMCL	ProtoAU	MAGCL
MovieLens-1M	Recall@10	0.1804	0.1657	0.1846	0.1830	0.1876	0.2011	0.1888	0.2045	0.2185	0.2088
	NDCG@10	0.2463	0.2295	0.2528	0.2510	0.2514	0.2644	0.2526	0.2636	0.2888	0.2674
	Recall@20	0.2714	0.2520	0.2741	0.2759	0.2796	0.2957	0.2848	0.3064	0.3192	0.3084
	NDCG@20	0.2569	0.2400	0.2614	0.2617	0.2620	0.2758	0.2649	0.2793	0.3021	0.2819
	Recall@50	0.4300	0.4122	0.4341	0.4364	0.4469	0.4653	0.4487	0.4758	0.4848	0.4780
	NDCG@50	0.3014	0.2851	0.3055	0.3056	0.3091	0.3240	0.3111	0.3283	0.3439	0.3315
Alibaba-iFashion	Recall@10	0.0303	0.0182	0.0382	0.0401	0.0457	0.0246	0.0461	0.0632	0.0488	0.0642
	NDCG@10	0.0161	0.0092	0.0198	0.0207	0.0246	0.0131	0.0248	0.0351	0.0266	0.0356
	Recall@20	0.0467	0.0302	0.0615	0.0634	0.0692	0.0038	0.0692	0.0917	0.0723	0.0925
	NDCG@20	0.0203	0.0123	0.0257	0.0266	0.0246	0.0165	0.0307	0.0423	0.0325	0.0428
	Recall@50	0.0799	0.0576	0.1081	0.1107	0.1144	0.0635	0.1141	0.1413	0.1181	0.1443
	NDCG@50	0.0269	0.0177	0.0349	0.0360	0.0396	0.0216	0.0396	0.0522	0.0421	0.0531

- **MultiGCCF** [9]: Constructs multiple graph structures to explicitly model user-item, user-user, and item-item relationships, and thus extracts richer information.
- **LightGCN** [8]: Simplifies traditional GCN by removing complex feature transformations and nonlinear activations, focusing instead on efficient neighbor aggregation.
- **SGL** [24]: Generates contrastive views via edge dropout and enhances performance using contrastive learning. The SGL-ED variant is used in this work.
- **SimGCL** [25]: Constructs contrastive views by adding random noise to the embedding space, improving both performance and training efficiency via contrastive learning.
- **AdaMCL** [36]: A GNN-based framework that adaptively models high-order relationships, leveraging contrastive learning to mine self-supervised signals and reduce noise through multi-task learning.
- **ProtoAU** [40]: Mitigates the bias caused by random sampling of contrastive pairs through prototype-based contrastive learning, and introduces alignment and uniformity objectives to prevent dimensional collapse.

Table 2 presents the comparison results of MAGCL against the nine baseline frameworks, with the best results highlighted in bold.

Several baseline models did not provide official source code or pre-trained implementations. Therefore, we adopted the reported evaluation

results directly from their original publications. Following common practice in recommender system research, all baseline results were obtained from experiments conducted on the same benchmark datasets (MovieLens-1M and Alibaba-iFashion) and evaluated using identical metrics (Recall@K and NDCG@K). For baselines with available implementations, we verified that their reported hyperparameter configurations, such as embedding size, batch size, and negative sampling strategy, were comparable to those used in our own experiments. This ensures that the comparisons are fair, consistent, and representative.

Table 2 shows that MAGCL achieves the best overall performance on Alibaba-iFashion and surpasses all baselines (BPRMF, NeuMF, NGCF, MultiGCCF, LightGCN, SGL, SimGCL, AdaMCL, and ProtoAU) across metrics. On MovieLens-1M, MAGCL is competitive and exceeds BPRMF, NeuMF, NGCF, MultiGCCF, LightGCN, SGL, SimGCL, and AdaMCL, while ProtoAU performs slightly better. The superior performance of ProtoAU can be attributed to its effective exploitation of the denser user-item interactions in MovieLens-1M, which enable the model to learn richer semantic representations. In contrast, MAGCL demonstrates stronger robustness under sparse data conditions by capturing high-order collaborative signals and mitigating the influence of noisy interactions. Specifically, compared with ProtoAU, MAGCL improves Recall@50 by 2.12%, NDCG@50 by 1.72%, and Recall@10 by 1.58% on the Alibaba-iFashion dataset. These results confirm that MAGCL achieves state-of-the-art performance, particularly in sparse and noisy recommendation scenarios.

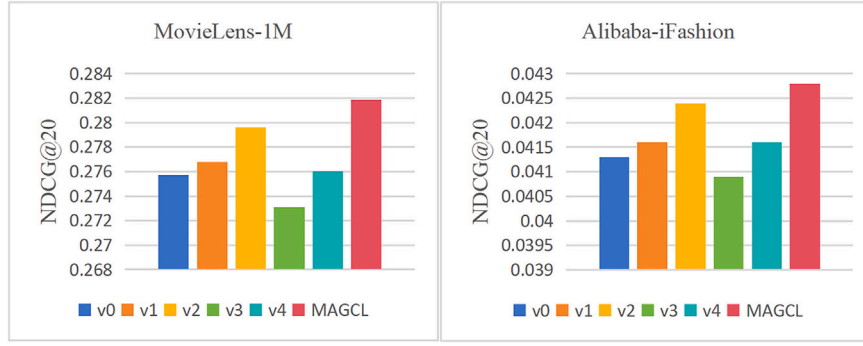


Fig. 6. NDCG@20 performance of MAGCL variants.

A comparative analysis of the frameworks reveals that GNN-based models consistently outperform traditional matrix factorization methods (e.g., BPRMF, NeuMF) across evaluation metrics, underscoring the advantage of graph structures in recommendation. NGCF outperforms early baselines, highlighting the efficacy of incorporating high-order collaborative signals. LightGCN further improves upon NGCF by eliminating feature transformation layers and nonlinear activations, suggesting that simplified architectures can enhance performance. Although MultiGCCF explicitly captures high-order relationships, its performance lags behind LightGCN, possibly due to residual noise and the continued use of linear transformations and nonlinear activations, which may hinder collaborative signal learning. SGL improves over LightGCN by constructing contrastive views through self-supervised learning, demonstrating the promise of contrastive learning in recommender systems. However, SGL falls short of MAGCL due to its inability to model high-order relationships, highlighting the necessity of capturing such signals. SimGCL, another contrastive method, performs poorly on Alibaba-iFashion, likely because of the injected random noise, which disturbs the inherent graph structure and leads to information loss. Finally, MAGCL significantly outperforms AdaMCL across both datasets, demonstrating the importance of effective noise reduction and embedding uniformity optimization in enhancing recommendation performance.

5.4. Ablation study

To evaluate the contribution of each major component of MAGCL more clearly, we conduct a set of ablation experiments. Specifically, five variants are designed under the same training and evaluation settings as the full model: v0 removes the denoising contrastive loss, v1 removes the high-order contrastive loss, v2 removes the uniformity loss, v3 removes the cross-layer contrastive loss, and v4 removes the dynamic fusion strategy, by replacing Eq. (13) with $z_x^l = z_x^{G,l} + z_x^{\hat{G},l}$, thereby simplifying dynamic fusion into a direct summation of node representations from the two graphs. Fig. 6 presents the NDCG@20 results for these variants on MovieLens-1M and Alibaba-iFashion.

The results show that MAGCL outperforms all variants, confirming that each component contributes positively. Specifically, v0 (removing denoising contrastive loss) reduces NDCG@20 by 2.20% and 3.50% on MovieLens-1M and Alibaba-iFashion, respectively, highlighting the effectiveness of constructing a denoised augmented graph in conjunction with contrastive learning. v1 (removing high-order contrastive loss) decreases NDCG@20 by 1.81% and 2.80%, underscoring the importance of high-order collaborative graph modeling with contrastive learning. v2 (removing uniformity loss) lowers NDCG@20 by 0.82% and 0.93%, suggesting that uniformity loss plays a key role in enhancing embedding quality and performance. Removing cross-layer contrastive loss (v3) reduces NDCG@20 by 3.12% and 4.44%, indicating that this loss helps suppress noise from high-order neighbor aggregation and improves recommendation quality. Removing any contrastive loss significantly degrades performance, emphasizing its critical role in MAGCL training.

Finally, v4 (removing dynamic fusion) decreases NDCG@20 by 2.1% and 2.8%, illustrating that dynamic fusion effectively models high-order relationships, thus enhancing recommendation effectiveness.

Overall, these ablation results verify that all components of MAGCL work together synergistically to improve robustness and recommendation accuracy.

6. Conclusion

This paper proposes MAGCL, a multi-view adaptive graph contrastive learning framework for robust recommendation. First, MAGCL captures high-order collaborative signals by constructing a high-order graph and employing a dynamic fusion strategy to enhance representation learning. Second, it introduces a multi-view contrastive learning scheme that incorporates the fused view, the denoised augmented graph, and the high-order collaborative graph, enabling more effective contrastive training and mitigating noise interference. Additionally, MAGCL integrates a cross-layer contrastive objective to reduce the impact of noisy high-order neighbors and optimizes a uniformity loss to further enhance recommendation accuracy. Experimental results on the MovieLens-1M and Alibaba-iFashion datasets demonstrate that MAGCL achieves significant improvements over representative baselines, including BPRMF, NeuMF, NGCF, MultiGCCF, LightGCN, SGL, SimGCL, AdaMCL, and ProtoAU (on Alibaba-iFashion), and surpasses all of the above except ProtoAU on MovieLens-1M.

In future work, we will further enhance MAGCL from two perspectives. First, we plan to explore lightweight data augmentation strategies to reduce training overhead while maintaining or further improving model performance. Second, we will evaluate MAGCL on additional large-scale implicit-feedback benchmarks to assess robustness and generalizability across domains and sparsity levels.

CRediT authorship contribution statement

Dongyu Jiang: Writing – original draft, Data curation. **Miao Liu:** Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization. **Wenbin Chen:** Writing – review & editing, Visualization, Resources, Investigation. **Ligang Zheng:** Visualization, Validation, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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